Warehouse Operations & Process Improvement Analysis Report

Objective
Identify where a warehouse loses processing time, why it may be happening, and what management should investigate first.

Dataset
10,000 simulated warehouse orders (August 2026), generated in Excel with deliberate patterns. Each order records its zone, shift, product category, items, priority, pick and pack times, errors, worker experience and shift workload.

Method
- Generated synthetic data in Excel with built-in patterns.
- Cleaned and validated it in Python (pandas): missing values, duplicates, data types and impossible values. Calculated KPIs and investigated zone, shift, workload and worker experience in Python, with six matplotlib charts.
- Cross-checked every result against Excel PivotTables, which matched. Built an Excel dashboard to
- present the findings.

Key results
12.1 min average processing time, 4.7% error rate, and 13.8% of orders delayed (over 15 minutes).

Findings
- Zone C is the main bottleneck: average processing time is 14.1 minutes, its error rate is 7.7%, and 33.8% of its own orders are delayed. Although it handles about 25% of all orders, it accounts for about 62% of all delayed orders.
- Evening is the slowest shift (12.8 min, 19.7% delayed). Zone C in the evening is the worst combination at 15.0 min.
- New workers are slower and less accurate (13.8 vs 11.4 min; 6.4% vs 3.9% errors), and slowest in Zone C (16.0 min).
- Delays rise with workload: 20.2% of orders are delayed in high-workload shifts against 10.0% in low-workload shifts.

These are patterns that warrant investigation, not proven causes. Workload and evening shifts overlap, so their effects cannot be separated with this data.

Recommendation
Prioritise an operational review of Zone C during peak evening periods, focusing on staffing allocation, product placement and new-worker support.
Suggested test: trial an additional experienced picker in Zone C during the busiest evening period for about two weeks, and compare processing time, delayed % and error rate against the current baseline.
Agree the success measure before starting, for example Zone C's delayed % moving towards the 13.8% overall rate.

Limitations
This is a simulated student dataset designed to demonstrate operational analysis. It does not represent actual Amazon data. The patterns were built into the data, so the project shows the analytical method rather than real warehouse behaviour. The 15-minute delay threshold is an assumption.